

14.8 METHOD OF COMPLEX AMPLITUDES

In place of using the sine or the cosine to represent a simple harmonic wave, one may write the equation in the exponential form*

$$y = ae^{i(\omega t - kx)} = ae^{i\omega t}e^{-i\delta}$$

where $\delta = kx$ and is constant at a particular point in space. The presence of $i = \sqrt{-1}$ in this equation makes the quantities complex. We can nevertheless use this representation and at the end of the problem take either the real (cosine) or the imaginary (sine) part of the resulting expression. The time-varying factor $\exp(i\omega t)$ is of no importance in combining waves of the same frequency, since the amplitudes and relative phases are independent of time. The other factor, $a \exp(-i\delta)$, is called the *complex amplitude*. It is a complex number whose modulus a is the real amplitude and whose argument δ gives the phase relative to some standard phase. The negative sign merely indicates that the phase is behind the standard phase. In general, the vector $\mathbf{a}$ is given by

$$\mathbf{a} = ae^{i\delta} = x + iy = a(\cos \delta + i \sin \delta)$$

Then it will be seen that

$$a = \sqrt{x^2 + y^2} \quad \tan \delta = \frac{y}{x}$$

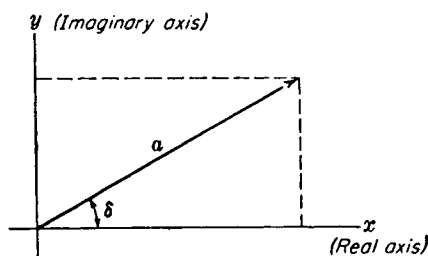
Thus if $\mathbf{a}$ is represented as in Fig. 14K, plotting horizontally its real part and vertically its imaginary part, it will have the magnitude a and will make the angle δ with the x axis, as we require for vector addition.

The advantage of using complex amplitudes lies in the fact that the algebraic

* For the mathematical background of this method, see E. T. Whittaker and G. N. Watson, "Modern Analysis," chap. 1, Cambridge University Press, New York, 1935.

FIGURE 14K

Representation of a vector in the complex plane.



addition of two or more is equivalent to vector addition of the real amplitudes. Thus for two such quantities

$$Ae^{i\theta} = a_1 e^{i\delta_1} + a_2 e^{i\delta_2}$$

so that if

$$x_1 + x_2 = a_1 \cos \delta_1 + a_2 \cos \delta_2 = X$$

and

$$y_1 + y_2 = a_1 \sin \delta_1 + a_2 \sin \delta_2 = Y$$

it will be found that our previous Eqs. (12d) and (12e) require that

$$A^2 = X^2 + Y^2 \quad \tan \theta = \frac{Y}{X} \quad (14l)$$

Thus, to get a vector sum, we need only obtain the *algebraic* sums $X = \Sigma x_i$ and $Y = \Sigma y_i$ of the real and imaginary parts, respectively, of the complex amplitudes. In obtaining the resultant intensity as proportional to the square of the real amplitude, we multiply the resultant complex amplitude by its complex conjugate, which is the same expression with i replaced by $-i$ throughout. The justification for this procedure follows from the relations

$$\begin{aligned} (X + iY)(X - iY) &= X^2 + Y^2 = A^2 \\ Ae^{i\theta} Ae^{-i\theta} &= A^2 \end{aligned} \quad (14m)$$

14.9 DERIVATION OF THE INTENSITY FUNCTION

For the fringe system formed by the transmitted light, the sum of the complex amplitudes is (see Fig. 14D)

$$\begin{aligned} Ae^{i\theta} &= att' + att'r^2 e^{i\delta} + att'r^4 e^{i2\delta} + \dots \\ &= a(1 - r^2)(1 + r^2 e^{i\delta} + r^4 e^{i2\delta} + \dots) \end{aligned}$$

where $1 - r^2$ has been substituted for tt' , according to Stokes' relation, Eq. (14c). The infinite geometric series in the second parentheses has the common ratio $r^2 \exp i\delta$, and has a finite sum because $r^2 < 1$. Summing the series, one obtains

$$Ae^{i\theta} = \frac{a(1 - r^2)}{1 - r^2 e^{i\delta}}$$

By Eq. (14m), the intensity is the product of this quantity by its complex conjugate, which yields

$$I_T \approx \frac{a(1-r^2) a(1-r^2)}{1-r^2 e^{i\delta} 1-r^2 e^{-i\delta}} = \frac{a^2(1-r^2)^2}{1-r^2(e^{i\delta} + e^{-i\delta}) + r^4}$$

Since $(e^{i\delta} + e^{-i\delta})/2 = \cos \delta$, and $a^2 \approx I_0$, the intensity of the incident beam, we obtain the result, in terms of real quantities only, as

$$I_T = I_0 \frac{(1-r^2)^2}{1-2r^2 \cos \delta + r^4} = \frac{I_0}{1 + [4r^2/(1-r^2)^2] \sin^2(\delta/2)} \quad (14n)$$

The main features of the intensity contours in Fig. 14I can be read from this equation. Thus at the maxima, where $\delta = 2\pi m$, we have $\sin^2(\delta/2) = 0$ and $I_T = I_0$. When the reflectance r^2 is large, approaching unity, the quantity $4r^2/(1-r^2)^2$ will also be large, and even a small departure of δ from its value for the maximum will result in a rapid drop of the intensity.

For the reflected fringes it is not necessary to carry through the summation, since we know from the conservation of energy that if no energy is lost through absorption,

$$I_R + I_T = 1 \quad (14o)$$

The reflected fringes are complementary to the transmitted ones, and for high reflectances become narrow dark fringes. These can be used to make more precise the study of the contour of surfaces.* If there is appreciable absorption on transmission through the surfaces, as will be the case if they are lightly silvered, one can no longer assume that Stokes' relations or Eq. (14o) hold. Going back to the derivation of Eq. (14n), it will be found that in this case the expression for I_T must be multiplied by $(t't')^2/(1-r^2)^2$. Here $t't'$ and r^2 are essentially the fractions of the intensity transmitted and reflected, respectively, by a single surface. Where the surfaces are metallized, there will be slight differences between t and t' , as well as small phase changes upon reflection. The transmitted fringes may still be represented by Eq. (14n), however, with an overall reduction of intensity and a correction to δ which merely changes slightly the effective thickness of the plate.